

Reider's Theorem, More and Less

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Let S be a smooth projective surface over $\mathbb{C}$.

1 Strategy

Going back to the ideas of [1], we will prove variants of Reider's theorem by finding suitable Bridgeland stability conditions.

Specifically, to show a line bundle $\omega_S(C)$ is d -very ample, we will get the vanishing

$$H^1(S, \mathcal{I}_Z \otimes \omega_S(C)) = 0,$$

as Z ranges over all length d subschemes. This cohomology group can be rewritten (with the help of Serre duality) as

$$\begin{aligned} \operatorname{Ext}_{\mathcal{O}_S}^1(\mathcal{O}_S(-C), \mathcal{I}_Z \otimes \omega_S) &= \operatorname{Ext}_{\mathcal{O}_S}^1(\mathcal{I}_Z, \mathcal{O}_S(-C))^\vee \\ &= \operatorname{Hom}_{D^b(S)}(\mathcal{I}_Z, \mathcal{O}_S(-C)[1])^\vee. \end{aligned}$$

To preclude any nonzero morphism $\mathcal{I}_Z \rightarrow \mathcal{O}_S(-C)[1]$ in the derived category, we will then find a Bridgeland stability condition with $\mathcal{I}_Z, \mathcal{O}_S(-C)[1]$ both (semi)stable but such that $\mathcal{I}_Z$ has greater slope.

2 Stability

2.1 Review

We use notation from [2], which includes a choice of smooth coordinates for a “slice” (of a chart) of the stability manifold. Specifically, we get a stability condition for each (x, y) inside the parabola $x > \frac{1}{2}y^2$, which is the data of a central charge $Z_{(x,y)}$ and a heart of a bounded t-structure $\mathcal{A}_{-y}$.

$$Z_{(x,y)}(E) = (-\operatorname{ch}_2(E), c_1(E) \cdot H) + (\operatorname{rk} E)(x, y)$$

$$\mathcal{A}_{-y} = \{E \in D^b(S) \mid -\mu_{\min}(\mathcal{H}^0(E)) < y \leq -\mu_{\max}(\mathcal{H}^{-1}(E)), \mathcal{H}^i(E) = 0 \text{ for } i \neq -1, 0\}$$

We recall here that as we increase y , we make the following trade:

- We *gain* H -semistable coherent sheaves with low H -slope, like $\mathcal{O}_S(-nC)$
- We *lose* shifts of H -semistable coherent sheaves with low H -slope, like $\mathcal{O}_S(-nC)[1]$.

In particular, an Mumford H -stable coherent sheaf $\mathcal{F}$ appears in the category when $y > -\mu(\mathcal{F})$.

Note that when $\mathcal{F}$ is *not* H -semistable, we have to wait a while after we lose $\mathcal{F}[1]$ before we gain $\mathcal{F}$. For instance, on $\mathbb{P}^2$ for $\mathcal{F} = \mathcal{O}_{\mathbb{P}^2}(-1) \oplus \mathcal{O}_{\mathbb{P}^2}(1)$, we have Harder-Narasimhan filtration

$$0 \leq \mathcal{O}_{\mathbb{P}^2}(1) \leq \mathcal{O}_{\mathbb{P}^2}(-1) \oplus \mathcal{O}_{\mathbb{P}^2}(1),$$

so has maximal H -slope $\mu_{\max}(\mathcal{F}) = 1$ and minimal slope $\mu_{\min}(\mathcal{F}) = -1$. Therefore, $\mathcal{F}[1]$ leaves the category once $y > -(1)$, but $\mathcal{F}$ only joins the category once $y > -(-1)$.

Also important is a description of (numerical) walls, which are lines through *points of indeterminacy* $I(E)$ of objects $E \in D^b(S)$, which are points $I(E) \in \mathbb{R}^2$ such that $Z_{I(E)}(E) = 0$. These have coordinates

$$I(E) = \left(\frac{\text{ch}_2 E}{\text{rk} E}, -\mu E \right).$$

Any H -semistable coherent sheaf $\mathcal{F}$ (or a shift, which has identical point of indeterminacy) has $I(\mathcal{F})$ outside or on the parabola—this is a version of the Bogomolov inequality. Alternatively, we can rephrase this argument as follows: if $\mathcal{F}$ is H -semistable, then $\mathcal{F} \in \mathcal{A}_{-\mu\mathcal{F}}$ so it is in the category at $I(\mathcal{F})$, so (x, y) cannot give a stability condition because the central charge sends an object to 0. Thus, (x, y) is outside the parabola.

When $\mathcal{F}$ fails to be semistable, like our example $\mathcal{F} = \mathcal{O}_{\mathbb{P}^2}(-1) \oplus \mathcal{O}_{\mathbb{P}^2}(1)$, we might have $I(\mathcal{F})$ inside the parabola, but it will occur at a y such that $\mathcal{F} \notin \mathcal{A}_{-y}$. For example, for this $\mathcal{F}$,

$$I(\mathcal{F}) = \left(\frac{\text{ch}_2(\mathcal{O}_{\mathbb{P}^2}(-1)) + \text{ch}_2(\mathcal{O}_{\mathbb{P}^2}(1))}{2}, \frac{-(-1) - (1)}{2} \right) = \left(\frac{1}{2}, 0 \right),$$

and indeed, $\mathcal{O}_{\mathbb{P}^2}(-1) \oplus \mathcal{O}_{\mathbb{P}^2}(1) \notin \mathcal{A}_0$ as discussed earlier.

2.2 $\mathcal{O}_S(-C)$ and Bogomolov-sharp Sheaves

Now, we analyze when $\mathcal{O}_S(-C)[1]$ is (semi)stable. First, we describe how, for an H -semistable torsion free coherent sheaf F , might F or $F[1]$ be destabilized. In particular, we need to know what a sub/quotient looks like in one of our hearts $\mathcal{A}_{-y}$.

Lemma 2.1. Let F be as above. Then,

- (1) If $K \leq F$ in $\mathcal{A}_{-y}$ with cokernel Q , then, by the long exact sequence in $\text{Coh}(S)$ cohomology,

$$0 \rightarrow \mathcal{H}^{-1}(K) \rightarrow 0 \rightarrow \mathcal{H}^{-1}(Q) \rightarrow \mathcal{H}^0(K) \rightarrow F \rightarrow \mathcal{H}^0(Q) \rightarrow 0$$

K is itself a sheaf with $\mu_{\min}(K) > -y$.

- (2) If $F[1] \twoheadrightarrow Q$ is a surjection in $\mathcal{A}_{-y}$ with kernel K , then, by the long exact sequence in $\text{Coh}(S)$ cohomology,

$$0 \rightarrow \mathcal{H}^{-1}(K) \rightarrow F \rightarrow \mathcal{H}^{-1}(Q) \rightarrow \mathcal{H}^0(K) \rightarrow 0 \rightarrow \mathcal{H}^0(Q) \rightarrow 0,$$

$Q[-1]$ is itself a sheaf with $\mu_{\max}(Q[-1]) \leq -y$.

Lemma 2.2. Let F be H -stable torsion free. Suppose there exists (x, y) in the parabola such that F is semistabilized by some proper subobject $E \xrightarrow{\varphi} F$ in $\mathcal{A}_{-y}$. Then, (x, y) is above $I(E)$ above $I(F)$, i.e. $y > -\mu(E) > -\mu(F)$.

Proof. First, assume $\text{rk} F > 0$, since otherwise the lemma is automatic. Continuing, since $E, F \in \mathcal{A}_{-y}$, we know

$$y > -\mu(F), -\mu_{\min}(E) \geq -\mu(E),$$

so at least (x, y) is above $I(E), I(F)$.

For $E \xrightarrow{\varphi} F$ to be a monomorphism in $\mathcal{A}_{-y}$, need the cone $C(\varphi) \in \mathcal{A}_{-y}$, which in particular means is the requirement that

$$\mu_{\max}(\mathcal{H}^{-1}(C(\varphi))) = \mu_{\max}(\ker_{\text{Coh}(S)} \varphi) \leq -y$$

Denote this kernel by K . If $K = 0$, then E injects in F in $\text{Coh}(S)$ and we win by H -stability of F . Otherwise, $K \neq 0$, is torsion free (by virtue of finite slope), so has positive rank. On the other extreme, we also cannot have $K = E$, since then

$$\mu(E) \leq -y, \quad \mu(E) > -y.$$

So, E/K is nonzero. Now, we have short exact sequence in $\text{Coh}(S)$

$$0 \rightarrow K \rightarrow E \rightarrow E/K \rightarrow 0,$$

which gives

$$H \cdot c_1(E) = H \cdot c_1(K) + H \cdot c_1(E/K), \quad (\star)$$

and since $E/K \leq F$ is a nonzero subobject in $\text{Coh}(S)$ with F torsion free, we know E/K is torsion free and

$$1 \leq \text{rk} E - \text{rk} K \leq \text{rk} F, \quad \mu(E/K) < \mu(F).$$

In particular, $\text{rk} E \geq 1$, so it has finite H -slope, and we can use $(\star)$ to solve for $\mu(E)$, giving us the equation

$$\mu(E) = \frac{\text{rk}(K)\mu(K)}{\text{rk} E} + \frac{\text{rk}(E/K)\mu(E/K)}{\text{rk} E}.$$

Plugging in our other inequalities, in particular using

$$\mu(K) \leq -y < \mu(F)$$

we get

$$\begin{aligned} \mu(E) &< \frac{\text{rk}(K)}{\text{rk}(E)}\mu(F) + \frac{\text{rk}(E) - \text{rk}(K)}{\text{rk}(E)}\mu(F) \\ &= \mu(F). \end{aligned}$$

□

Lemma 2.3. With the same assumption as lemma 2.2, but also assuming E is (x, y) -semistable, $I(E)$ is outside the parabola and on the same side of the parabola as $I(F)$.

Proof. $I(E)$ is outside of the parabola by a variant of the Bogomolov inequality for (x, y) -semistable objects (add reference/collect Bayer-Macri-Toda theorem here too). Then (x, y) above $I(E)$ above $I(F)$ as in lemma 2.2 implies the result. $\square$

Lemma 2.4. Let F be an H -semistable torsion free sheaf. Then, F is $(x, -\mu(F) + \varepsilon)$ -semistable for $x \gg 0$ and $0 < \varepsilon \ll 1$.

Proof. prove or cite this!! probably also need more in my applications to be careful... $\square$

Now, we apply these lemma to get the stability of $\mathcal{O}_S(-C)[1]$ and friends, using the observation that

$$I(\mathcal{O}_S(-C)[1]) = \left(\frac{C^2}{2}, \|C\| \right)$$

is exactly on the parabola. In this sense, $\mathcal{O}_S(-C)[1]$ is *Bogomolov-sharp* (the inequality became an equality), and we will show such Bogomolov-sharp object are stable in the following sense.

Proposition 2.5. Let $F \in \text{Coh}(S)$ be H -stable and torsion free such that $I(F)$ lands on the parabola $x = \frac{1}{2}y^2$. Then,

- (1) F is (x, y) semistable for all (x, y) in the parabola with $y > -\mu(F)$.
- (2) $F[1]$ is (x, y) semistable for all (x, y) in the parabola with $y \leq -\mu(F)$.

Proof. We show (1). By H -stability, F will be $(x, -\mu(F) + \varepsilon)$ -semistable for $x \gg 0$ and $0 < \varepsilon \ll 1$ as in lemma 2.4, and F will *remain* (x, y) -semistable until just after a wall, which is a line through $I(F)$. Let (x, y) be on the first destabilizing wall (maybe define this by saying that E destabilizes F in all neighborhoods of the wall?) counterclockwise to the positive horizontal ray from $I(F)$, and let

$$0 \rightarrow E \rightarrow F \rightarrow F/E \rightarrow 0$$

for $E, F/E \in \mathcal{A}_{-y}$ be a *witness* for this wall, where we choose E to be (x, y) semistable, i.e., by choosing the first factor in the (x, y) Harder-Narasimhan filtration for F . By lemma 2.2 and 2.3, $I(F), I(E), (x, y)$ form a positive slope line with increasing heights in order with $I(E)$ outside the parabola, but this is impossible because the line segment $I(F)$ to (x, y) is entirely inside the parabola. Thus, $I(F)$ couldn't have been destabilized. $\square$

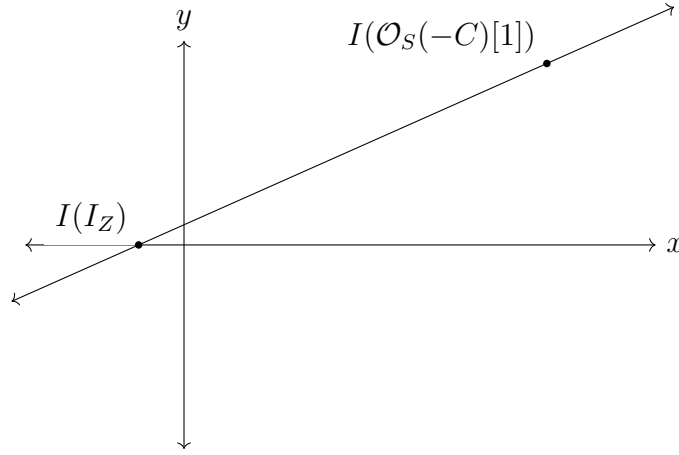
TODO: show (2) in a parallel way... might need to expand the lemmas though.

2.3 Ideal Sheaves

Unlike $\mathcal{O}_S(-C)[1]$, an ideal sheaf $\mathcal{I}_Z$ for a length d subscheme will not always be stable. However, we will establish numerical criteria to guarantee stability of the $\mathcal{I}_Z$ long enough to find (x, y) where $\mathcal{I}_Z$ has bigger (x, y) slope than $\mathcal{O}_S(-C)[1]$.

By openness of stability, we just need to guarantee $\mathcal{I}_Z$ is stable at a point on the line through $I(\mathcal{I}_Z)$ and $I(\mathcal{O}_S(-C)[1])$ intersected with the parabola. By lemma 2.4, since $\mathcal{I}_Z$ is H -stable torsion free, we at least know $\mathcal{I}_Z$ is semistable at $(x, -\mu(F) + \varepsilon)$ for $x \gg 0$ and $0 < \varepsilon \ll 1$, i.e., $\mathcal{I}_Z$ is stable in some circular sector bounded by the positive x -axis and some positive slope ray.

so we just need to show that F remains stable through the ray from $I(\mathcal{I}_Z)$ to $I(\mathcal{O}_S(-C)[1])$.



Now, if (x, y) is on some destabilizing wall, then in a neighborhood F has to be destabilized by a sheaf F , which we can assume to be (x, y) -semistable. We will refer to $\text{rk} F$ as the *rank of the wall* for $\mathcal{I}_Z$ through $I(F)$. We want to establish numerical conditions on (S, C) to prevent any walls that occur up to the $\mathcal{I}_Z, \mathcal{O}_S(-C)[1]$ wall. Our target wall has slope (in our coordinates for $\text{Stab}(S)$)

$$m = \frac{\|C\|}{\frac{C^2}{2} + d} = \frac{2\|C\|}{C^2 + 2d},$$

so we want to guarantee that any other wall through $\mathcal{I}_Z$ has greater slope.

2.3.1 High Rank Walls

Suppose $\text{rk} F = r > 1$. As in [2], we show that this implies the wall W has too great a “height”, i.e., showing that the ratio of the y coordinates of the top intersection and bottom intersection of $W \cap \{x = \frac{1}{2}y^2\}$ is too large.

For comparison, we compute this height ratio for the $\mathcal{I}_Z, \mathcal{O}_S(-C)[1]$ wall. The top intersection has y coordinate $\|C\|$. The bottom intersection has is one of the solutions to the equations

$$2x = y^2, \quad y = m(x + d),$$

which has y -coordinate which is a root of

$$y = m \left(\frac{1}{2} y^2 \right) + md$$

$$0 = y^2 - \frac{2}{m} y + 2d.$$

Since one root is $\|C\|$, the other root is $\frac{2d}{\|C\|}$. Thus, the height ratio is $\frac{C^2}{2d}$ (or potentially the reciprocal if $\|C\|$ isn't sufficiently large—but we will eventually require $\|C\|$ to be large enough so that $\|C\|$ is the bigger height).

Now, we bound the height ratio for a rank r wall. By Lemma 2.3, $I(F)$ is positioned between $I(I_Z)$ and the parabola, with positive y -value. Writing E for the cone, so that we have exact sequence

$$0 \rightarrow F \rightarrow I_Z \rightarrow E \rightarrow 0 \quad (\star)$$

in $\mathcal{A}_{-y}$ (where (x, y) is on the destabilizing wall). Since F, I_Z are both (x, y) -semistable of the same slope, so is E (**exercise**). Thus, $I(E)$ is also outside of the parabola, but the invariants of E will force $I(E)$ to be on the *opposite* side, since $(\star)$ implies

$$\mu(E) = \frac{r}{r-1} \mu(F), \quad \text{ch}_2(E) = -d - \text{ch}_2(F)$$

so... **uh oh, why opposite side? for $r \gg 0$, this is basically the same height, so I'm concerned.**

Thus, the heights of $I(F)$, $I(E)$ give lower and upper bounds respectively on the height ratio for this wall

TODO:

- understand “large volume limit”
- understand H -stability of $\mathcal{O}_S$ and I_Z .
- replace equations labeled with $\star$ with numbered equations.

References

- [1] Daniele Arcara and Aaron Bertram. *Reider's Theorem and Thaddeus Pairs Revisited*. Apr. 22, 2009. DOI: 10.48550/arXiv.0904.3500. arXiv: 0904.3500[math]. URL: <http://arxiv.org/abs/0904.3500> (visited on 02/07/2026).
- [2] Aaron Bertram et al. *Two New Extensions of Reider's Theorem on Algebraic Surfaces*. Apr. 23, 2026. DOI: 10.48550/arXiv.2604.21242. arXiv: 2604.21242[math]. URL: <http://arxiv.org/abs/2604.21242> (visited on 04/24/2026).